

DSA0404- Fundamentals of Data Science

CO3-AT1

Case Study 1 — Detecting an Unusual Customer

Given Data

- Average amount spent (Mean, μ) = ₹2,000
- Standard Deviation (σ) = ₹400
- Customer's Spending (x) = ₹3,200

A. Calculate the customer's z-score.

The z-score indicates how many standard deviations a data value is away from the mean.

Formula

$$z = \frac{x - \mu}{\sigma}$$

where

- x = observed value = ₹3,200
- μ = mean = ₹2,000
- σ = standard deviation = ₹400
-

Step 1: Find the difference between the observed value and the mean

$$\begin{aligned} x - \mu &= 3200 - 2000 = \\ &= 1200 \end{aligned}$$

This means the customer spends **₹1,200 more than the average customer.**

Step 2: Divide by the standard deviation

$$z = \frac{1200}{400}$$

$$z = 3$$

$$z = 3$$

The customer's spending is **3 standard deviations above the average**.

B. Is the customer's spending unusually high compared with the average?

A common rule for interpreting z-scores is:

Z-score	Interpretation
0	Exactly average
±1	Close to average
±2	Unusual
±3 or more	Very unusual (extreme value)

The calculated z-score is

$$z = 3$$

This means the customer's spending is three standard deviations above the mean.

Since values greater than +3 are considered rare and unusual, this customer's spending is significantly higher than normal.

According to the empirical (68–95–99.7) rule:

- About 68% of customers lie within $\pm 1\sigma$.
- About 95% lie within $\pm 2\sigma$.
- About 99.7% lie within $\pm 3\sigma$.
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Therefore, a customer at $z = +3$ is among the highest-spending customers, with only about 0.15% of customers expected to spend more on the upper tail of a normal distribution. Yes. Since the z-score is +3, the customer's spending is unusually high compared with the average and represents an extreme observation.

C. What would a negative z-score mean in this context?

A negative z-score indicates that the customer's spending is less than the average spending.

This happens because

$$x < \mu$$

making

$$x - \mu < 0$$

and therefore the z-score becomes negative.

Interpretation

A negative z-score tells us that:

- The customer spends less than the average customer.
- The larger the negative value, the lower the spending.
- Very negative values (such as -3) indicate extremely low spending compared with most customers.

A negative z-score means the customer's expenditure is below the average spending level. The more negative the score, the lower the spending relative to other customers.

D. If another customer has a z-score of -1.5 , what does it tell you?

Given:

$$z = -1.5$$

This means the customer's spending is 1.5 standard deviations below the mean.

Step 1: Find the actual spending

Using the z-score formula:

$$z = \frac{x - \mu}{\sigma}$$

Substitute the known values:

$$-1.5 = \frac{x - 2000}{400}$$

Multiply both sides by 400:

$$\begin{aligned} -1.5 \times 400 &= x - 2000 \\ -600 &= x - 2000 \end{aligned}$$

Add 2000 to both sides:

$$\begin{aligned} x &= 2000 - 600 \\ x &= 1400 \end{aligned}$$

Verification

$$\begin{aligned} z &= \frac{1400 - 2000}{400} \\ &= \frac{-600}{400} \\ &= -1.5 \end{aligned}$$

The calculation is correct.

Interpretation

A customer with a z-score of -1.5 :

- Spends ₹1,400.
- Spends ₹600 less than the average customer.
- Is below average, but not an extreme outlier.
- Falls within the normal range of customer spending, although on the lower side.

A z-score of -1.5 indicates that the customer spends **1.5 standard deviations below the average**, corresponding to an estimated spending of **₹1,400**. This customer spends less than the typical customer but is not considered an unusually low spender.

Case Study 2 – Delivery Time Analysis

Given Data

The delivery times (in minutes) recorded by the company are:

25	27	28	29	30	30	31	32	34	60
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Number of observations (n) = 10

The data is already arranged in **ascending order**, making it easier to calculate the measures of central tendency and dispersion.

1. Calculate the Mean and Median

(A) Mean (Arithmetic Average)

The mean is the average of all observations.

$$\bar{x} = \frac{2 + 3 + 7}{3} = \frac{12}{3} = 4$$

$\bar{x}$ is the arithmetic mean (simple average)

Formula

$$\text{Mean} = \frac{\sum x}{n}$$

where

- $\sum x$ = Sum of all observations
- n = Number of observations

Step 1: Calculate the Sum

Delivery Time	Running Total
25	25
27	52
28	80
29	109
30	139
30	169
31	200
32	232
34	266
60	326

Therefore,

$$\sum x = 326$$

Step 2: Calculate the Mean

$$\begin{aligned}\text{Mean} &= \frac{326}{10} \\ &= 32.6 \text{ minutes}\end{aligned}$$

Mean = 32.6 minutes

(B) Median

The **median** is the middle value when the data is arranged in ascending order.

Since

$$n = 10$$

which is an **even number**, the median is the average of the **5th and 6th observations**.

Locate the Middle Values

Position	1	2	3	4	5	6	7	8	9	10
Value	25	27	28	29	30	30	31	32	34	60

The middle observations are

5th = 30

6th = 30

Calculate the Median

$$\begin{aligned}\text{Median} &= \frac{30 + 30}{2} \\ &= \frac{60}{2} \\ &= 30\end{aligned}$$

Median = 30 minutes

Observation

Although one delivery took **60 minutes**, the median remained **30 minutes**, indicating that it is **not affected by extreme values**.

B. Find Q_1 and Q_3

Quartiles divide the data into four equal parts.

Step 1: Divide the Dataset

Lower Half

25, 27, 28, 29, 30

Upper Half

30, 31, 32, 34, 60

First Quartile (Q_1)

Q_1 is the median of the lower half.

Lower half:

25, 27, **28**, 29, 30

The middle value is

$$Q_1 = 28$$

Third Quartile (Q_3)

Upper half:

30, 31, **32**, 34, 60

The middle value is

$$Q_3 = 32$$

Answer

$$Q_1 = 28$$

$$Q_3 = 32$$

Interpretation

- 25% of deliveries take **28 minutes or less**.
- 75% of deliveries take **32 minutes or less**.

C. Calculate the Interquartile Range (IQR)

The **Interquartile Range (IQR)** measures the spread of the middle **50%** of the data.

Formula

$$IQR = Q_3 - Q_1$$

$$IQR = 32 - 28$$

$$IQR = 4$$

$$IQR = 4 \text{ minutes}$$

Interpretation

The middle 50% of delivery times vary only by 4 minutes, indicating that most deliveries are fairly consistent.

D. Check for Outliers Using the $1.5 \times IQR$ Rule

An outlier is an observation that is unusually far from the rest of the data.

Step 1: Calculate $1.5 \times IQR$

$$\begin{aligned} 1.5 \times IQR \\ &= 1.5 \times 4 \\ &= 6 \end{aligned}$$

Step 2: Calculate the Lower Limit

$$\begin{aligned} \text{Lower Limit} &= Q_1 - 1.5(IQR) \\ &= 28 - 6 \\ &= 22 \end{aligned}$$

Step 3: Calculate the Upper Limit

$$\begin{aligned} \text{Upper Limit} &= Q_3 + 1.5(IQR) \\ &= 32 + 6 \\ &= 38 \end{aligned}$$

Step 4: Compare Every Observation

Delivery Time	Within Range?
25	Yes
27	Yes
28	Yes

Delivery Time	Within Range?
29	Yes
30	Yes
30	Yes
31	Yes
32	Yes
34	Yes
60	No

Acceptable range:

$$22 \leq x \leq 38$$

Since

$$60 > 38$$

it lies outside the acceptable range.

60 minutes is an outlier.

Interpretation

The 60-minute delivery is significantly longer than the remaining delivery times. It may have occurred due to:

- Heavy traffic
- Vehicle breakdown
- Incorrect address
- Bad weather
- Unexpected delivery delays

The company should investigate this delivery separately rather than considering it representative of normal performance.

E. Explain Whether the Mean or Median Gives a Better Description of the Typical Delivery Time

Normally, the **mean** is preferred because it uses all observations.

However, the dataset contains an **extreme value (60 minutes)**.

This unusually high value increases the mean from around **30 minutes** to **32.6 minutes**, making the average appear larger than what most deliveries actually take.

The **median** is unaffected by this outlier because it depends only on the middle observations.

Hence, the median better represents the **typical delivery time**.

Comparison

Mean	Median
Affected by outliers	Not affected by outliers
Uses every observation	Uses only the middle observation(s)
32.6 minutes	30 minutes
Less representative here	More representative here

Conclusion

Since the dataset contains an outlier (60 minutes), the median (30 minutes) provides a more accurate description of the typical delivery time than the mean (32.6 minutes).

Case Study 3 – Comparing Temperature Variability Between Two Cities

Given Data

The average daily temperatures (°C) recorded over **7 days** are:

Day	1	2	3	4	5	6	7
City A (°C)	28	29	30	30	31	30	32
City B (°C)	20	25	30	35	30	25	45

Number of observations,

$$n = 7$$

A. Calculate the Mean Temperature for Each City

The **mean** is the arithmetic average of all observations.

Formula

$$\text{Mean} = \frac{\sum X}{n}$$

where

- $\sum X$ = Sum of all observations
- n = Number of observations

City A

Step 1: Calculate the Sum

$$\begin{aligned} & 28 + 29 + 30 + 30 + 31 + 30 + 32 \\ & = 210 \end{aligned}$$

Step 2: Calculate the Mean

$$\begin{aligned}\text{Mean} &= \frac{210}{7} \\ &= 30^{\circ}\text{C}\end{aligned}$$

$$\boxed{\text{Mean of City A} = 30^{\circ}\text{C}}$$

City B

Step 1: Calculate the Sum

$$\begin{aligned}20 + 25 + 30 + 35 + 30 + 25 + 45 \\ = 210\end{aligned}$$

Step 2: Calculate the Mean

$$\begin{aligned}\text{Mean} &= \frac{210}{7} \\ &= 30^{\circ}\text{C}\end{aligned}$$

$$\boxed{\text{Mean of City B} = 30^{\circ}\text{C}}$$

B. Calculate the Variance for Each City

The **variance** measures how far each observation is spread from the mean.

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

Formula

$$\text{Variance} = \frac{\sum (X - \mu)^2}{n}$$

where

- X = Observation
- μ = Mean

- n = Number of observations

Variance of City A

Mean = 30°C

Temperature (X)	Mean (μ)	Deviation (X- μ)	Square of Deviation (X- μ) ²
28	30	-2	4
29	30	-1	1
30	30	0	0
30	30	0	0
31	30	1	1
30	30	0	0
32	30	2	4
Total			10

$$\text{Variance} = \frac{10}{7}$$

$$= 1.4286$$

Variance of City A = 1.43

Variance of City B

Mean = 30°C

Temperature (X)	Mean (μ)	Deviation (X- μ)	Square of Deviation (X- μ) ²
20	30	-10	100
25	30	-5	25

Temperature (X)	Mean (μ)	Deviation ($X-\mu$)	Square of Deviation ($X-\mu$) ²
30	30	0	0
35	30	5	25
30	30	0	0
25	30	-5	25
45	30	15	225
Total			400

$$\text{Variance} = \frac{400}{7}$$

$$= 57.14$$

Variance of City B = 57.14

Observation

- City A has a **very small variance (1.43)**.
- City B has a **much larger variance (57.14)**.

This indicates that temperatures in City B fluctuate much more than those in City A.

C. Which City Has Greater Temperature Variability?

Temperature variability is determined by comparing the variances.

City	Variance
City A	1.43
City B	57.14

Since

$$57.14 > 1.43$$

City B has much greater temperature variation.

D. Why Can Two Datasets Have the Same Mean but Different Standard Deviations?

The **mean** indicates the central value of a dataset, whereas the **standard deviation (or variance)** measures how spread out the data is around the mean. Although both City A and City B have the same mean of **30°C**, their temperature distributions are very different.

City A

28, 29, 30, 30, 31, 30, 32

- Temperatures are clustered close to 30°C.
- Deviations from the mean are very small.
- Hence, the variance and standard deviation are low.

City B

20, 25, 30, 35, 30, 25, 45

- Temperatures are spread over a much wider range.
- Deviations from the mean are much larger.
- Hence, the variance and standard deviation are high.

Therefore

Two datasets can have the **same mean** because their total values average to the same number, but they can have **different standard deviations** because the individual observations are distributed differently around that mean.

E. Which City Has More Predictable Temperatures?

A city with **lower variance** has temperatures that remain closer to the average and are therefore more predictable.

From the calculations,

- Variance of City A = **1.43**
- Variance of City B = **57.14**

Since City A has a much lower variance, its temperatures change very little from day to day.

City B experiences much larger fluctuations, making future temperatures less predictable.

Case Study 4 – CPU Usage Analysis

Given Data

The CPU usage percentages recorded are:

40	42	45	45	46	48	50	52	55	90
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The data is already arranged in **ascending order**.

$$n = 10$$

A. Calculate the Mean, Median and Mode

Mean

$$\bar{x} = \frac{2 + 4 + 7}{3} = \frac{13}{3} = 4 \frac{1}{3}$$

$$\bar{X} = \frac{\sum X}{n}$$

The sum of all CPU usage values is

$$40 + 42 + 45 + 45 + 46 + 48 + 50 + 52 + 55 + 90 \\ = 513$$

Therefore,

$$\bar{X} = \frac{513}{10} \\ = 51.3\%$$

Median

Since there are **10 observations (even number)**, the median is the average of the **5th and 6th observations**.

Position	1	2	3	4	5	6	7	8	9	10
Value	40	42	45	45	46	48	50	52	55	90

The middle values are

46 and 48

Therefore,

$$\begin{aligned}\text{Median} &= \frac{46 + 48}{2} \\ &= \frac{94}{2} \\ &= \mathbf{47\%}\end{aligned}$$

Mode

The mode is the value that occurs most frequently.

CPU Usage	Frequency
40	1
42	1
45	2
46	1
48	1
50	1
52	1
55	1
90	1

Since **45** appears **twice**, while every other value appears only once,

$$\boxed{\text{Mode} = 45}$$

B. Calculate the Range

The range measures the difference between the highest and lowest observations.

The formula is

$$\text{Range} = \text{Maximum Value} - \text{Minimum Value}$$

The highest CPU usage is

90

The lowest CPU usage is

40

Therefore,

$$\begin{aligned}\text{Range} &= 90 - 40 \\ &= 50\end{aligned}$$

Hence,

$\text{Range} = 50$

C. Find Q_1 , Q_3 and IQR

The observations are divided into two equal halves.

Lower Half

40, 42, 45, 45, 46

The middle observation is

$$Q_1 = 45$$

Upper Half

48, 50, 52, 55, 90

The middle observation is

$$Q_3 = 52$$

The Interquartile Range (IQR) is

$$\begin{aligned} IQR &= Q_3 - Q_1 \\ &= 52 - 45 \\ &= 7 \end{aligned}$$

Hence,

$$\boxed{Q_1 = 45}$$

$$\boxed{Q_3 = 52}$$

$$\boxed{IQR = 7}$$

D. Identify Possible Outliers

The **1.5 × IQR Rule** is used to identify outliers.

First,

$$\begin{aligned} 1.5 \times IQR \\ &= 1.5 \times 7 \\ &= 10.5 \end{aligned}$$

The lower limit is

$$\begin{aligned} Q_1 - 1.5(IQR) \\ &= 45 - 10.5 \\ &= 34.5 \end{aligned}$$

The upper limit is

$$\begin{aligned} Q_3 + 1.5(IQR) \\ &= 52 + 10.5 \\ &= 62.5 \end{aligned}$$

Therefore,

CPU Usage	Within Limits?
40	Yes
42	Yes
45	Yes
45	Yes
46	Yes
48	Yes
50	Yes
52	Yes
55	Yes
90	No

The acceptable range is

$$34.5 \leq X \leq 62.5$$

Since

$$90 > 62.5$$

the value **90%** is an outlier.

Outlier = 90%

E. Determine the Direction of Skewness

The direction of skewness can be determined by comparing the **mean**, **median**, and **mode**.

Measure Value

Mean 51.3

Median 47

Mode 45

Since

$$\text{Mean} > \text{Median} > \text{Mode}$$

the distribution is **positively skewed (right-skewed)**.

The unusually high CPU usage of **90%** stretches the distribution toward the right side, increasing the mean more than the median.

Therefore, the dataset exhibits **positive skewness**.

The distribution is positively (right) skewed.
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Case Study 5– Comparing Two Datasets Using Mean, Variance and Standard Deviation

Given Data

Dataset A

10, 20, 30, 40, 50

Dataset B

28, 29, 30, 31, 32

Number of observations,

$$n = 5$$

A. Calculate the Mean of Both Datasets

The arithmetic mean is calculated using

$$\bar{X} = \frac{\sum X}{n}$$

Dataset A

$$\begin{aligned}\sum X &= 10 + 20 + 30 + 40 + 50 \\ &= 150 \\ \bar{X} &= \frac{150}{5} \\ &= 30\end{aligned}$$

Dataset B

$$\begin{aligned}\sum X &= 28 + 29 + 30 + 31 + 32 \\ &= 150 \\ \bar{X} &= \frac{150}{5} \\ &= 30\end{aligned}$$

Both datasets have the same mean.

Mean of Dataset A = 30

Mean of Dataset B = 30

B. Calculate the Variance of Both Datasets

The variance is calculated using

$$\sigma^2 = \frac{\sum (X - \mu)^2}{n}$$

where

- X = Observation
- μ = Mean
- n = Number of observations

Dataset A

Mean,

$$\mu = 30$$

Observation (X)	Mean (μ)	$X - \mu$	$(X - \mu)^2$
10	30	-20	400
20	30	-10	100
30	30	0	0
40	30	10	100
50	30	20	400
Σ			1000

Therefore,

$$\begin{aligned}\sigma^2 &= \frac{1000}{5} \\ &= 200\end{aligned}$$

Dataset B

Mean,

$$\mu = 30$$

Observation (X)	Mean (μ)	$X - \mu$	$(X - \mu)^2$
28	30	-2	4
29	30	-1	1
30	30	0	0
31	30	1	1
32	30	2	4
Σ			10

Therefore,

$$\begin{aligned}\sigma^2 &= \frac{10}{5} \\ &= 2\end{aligned}$$

Hence,

Variance of Dataset A = 200

Variance of Dataset B = 2

C. Calculate the Standard Deviation of Both Datasets

The standard deviation is the square root of the variance.

$$\sigma = \sqrt{\sigma^2}$$

Dataset A

Variance

$$= 200$$

Therefore,

$$\begin{aligned}\sigma &= \sqrt{200} \\ &= \sqrt{100 \times 2} \\ &= 10\sqrt{2} \\ &= 10 \times 1.414 \\ &= 14.14\end{aligned}$$

Dataset B

Variance = 2

Therefore,

$$\begin{aligned}\sigma &= \sqrt{2} \\ &= 1.414\end{aligned}$$

Hence,

Standard Deviation of Dataset A = 14.14

Standard Deviation of Dataset B = 1.41

D. Which Dataset Has Greater Spread?

The spread of a dataset is measured using the **variance** or **standard deviation**.

Dataset	Variance	Standard Deviation
A	200	14.14
B	2	1.41

Dataset A has much larger values for both variance and standard deviation.

This indicates that its observations are spread much farther from the mean.

Dataset B has observations very close to the mean.

Therefore,

Dataset A has the greater spread.

E. If Both Datasets Represent Model Errors, Which Model Appears More Consistent?

A model with **smaller variance** and **smaller standard deviation** produces errors that are more consistent.

Dataset B has

- Variance = 2
- Standard Deviation = 1.41

whereas Dataset A has

- Variance = 200
- Standard Deviation = 14.14

The values in Dataset B remain very close to the mean error, indicating stable and reliable model performance.

Dataset A shows much larger fluctuations in its errors.

Therefore,

The model represented by Dataset B is more consistent.